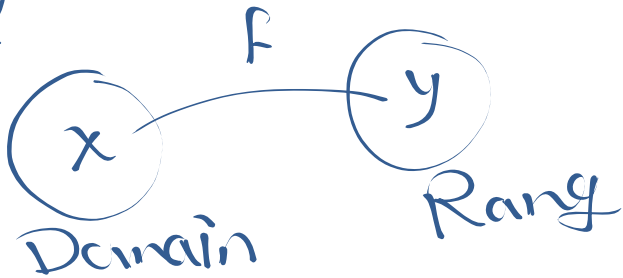
Function

$y = f(x)$



$$y = a_1 x + a_2 x^2 + a_3 x^3 + \cdots$$

$y = 3x^2 + 2x + 1$

Domain $= R =]-\infty, \infty[$

Range $= R$

Radical function

$y = \sqrt{x+5}$ $\quad x + 5 \geq 0$

$x \geq -5$ $\quad$ Domain $= [-5, \infty[$

$y = \sqrt[3]{x+5}$ $\quad y = \sqrt[4]{\ }$

$\sqrt[3]{-8} = -2$ $\quad$ Domain $= R$

$y = \sqrt{x+1}$

$x + 1 \geq 0$

$x \geq -1$

Domain $= [-1, \infty[$

$y = \sqrt[3]{x+1}$

Domain $= \mathbb{R}$

Rational Function

$y = \frac{x^2+1}{x-5} \Rightarrow$ Domain $= \mathbb{R} - \{5\}$

x — 3 hr

y — 6 hr

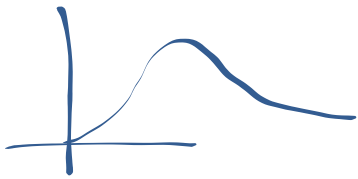
$\Rightarrow \frac{1}{t} = \frac{1}{3} + \frac{1}{6} = \frac{1}{2}$

$t = 2$

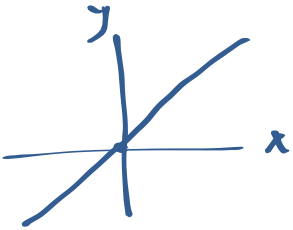
$\frac{1}{t} = \frac{1}{3} + \frac{1}{6}$

$v = \sqrt{C d}$ m/s

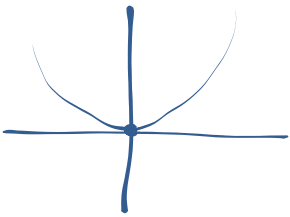
$C = \frac{4t}{t^2+1}$



$y = x$



$y = x^2$



Domain $= \mathbb{R}$
Range $= [0, \infty[$

$y = x^3$



Domain $= \mathbb{R}$
Range $= \mathbb{R}$

$y = \sqrt{x}$



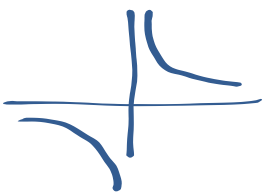
Range Domain $= [0, \infty[$

$y = \sqrt[3]{x}$

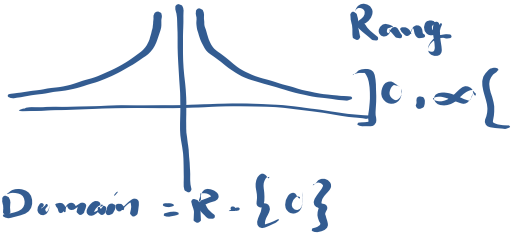


$D = \text{Range} = \mathbb{R}$

$y = \frac{1}{x}$



$y = \frac{1}{x^2}$

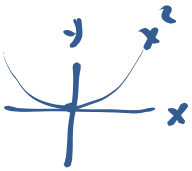


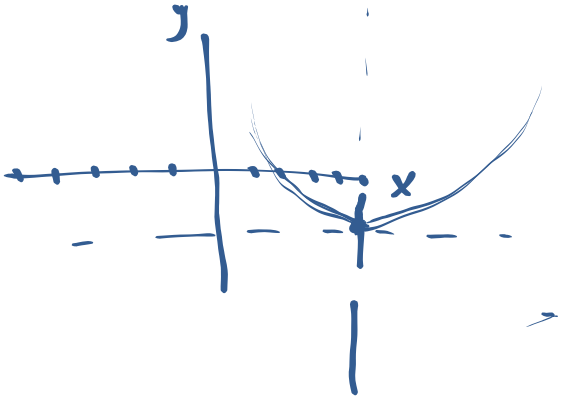
$y \pm h = f(x \pm k)$

$(\pm k, \pm h)$

$(y+1) = (x-5)^2$

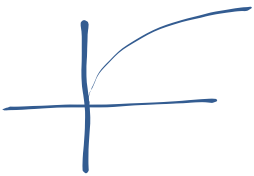
$(5, -1)$

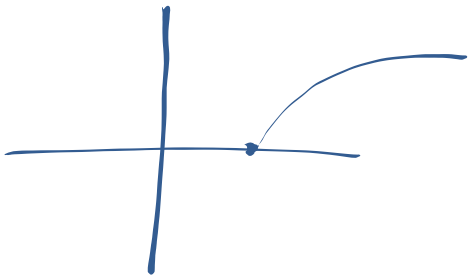




$y = \sqrt{x-1}$

$(1, 0)$

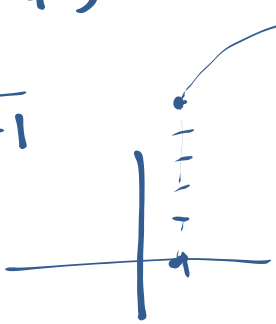




$y = \sqrt{x-1} + 5$

$y - 5 = \sqrt{x-1}$

$(1, 5)$



Domain $= [1, \infty[$

Range $= [5, \infty[$

$y = f(x)$

$y = \sqrt{-x}$



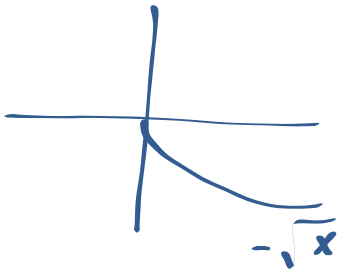


Domain $]-\infty, 0]$

Range $= [0, \infty[$

$y = -\sqrt{x}$

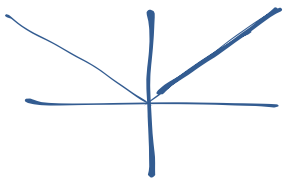




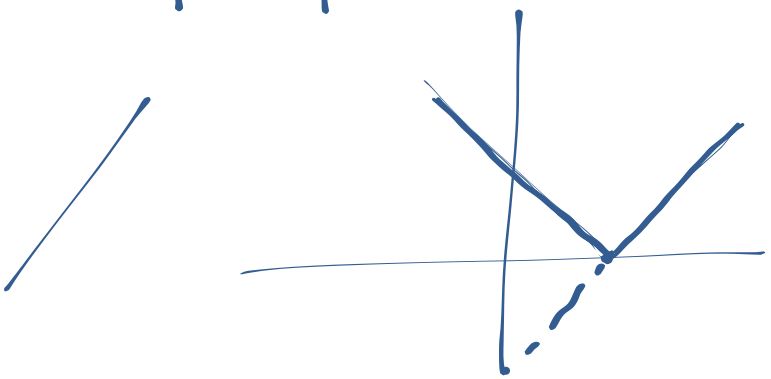
$y = x$



$y = |x|$



$y = |x - 1|$ $(1, 0)$



Combination of Function

$f(x) = \sqrt{x}$ $g(x) = x + 1$

$f + g = \sqrt{x} + x + 1$ $D_f \cap D_g$

$D_f = [0, \infty[$ $D_g =]-\infty, \infty[$

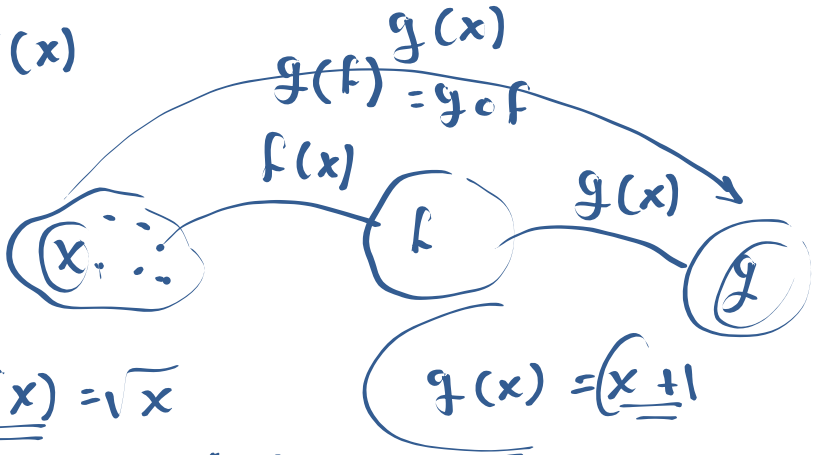
$D_f \cap D_g = [0, \infty[$

$f - g$ $D_f \cap D_g$

$f * g$ " " "

$\frac{f}{g} = \frac{\sqrt{x}}{x+1}$ $[0, \infty[- \{-1\}$

Composition of Function



$f(x) = \sqrt{x}$ $\quad$ $g(x) = x + 1$

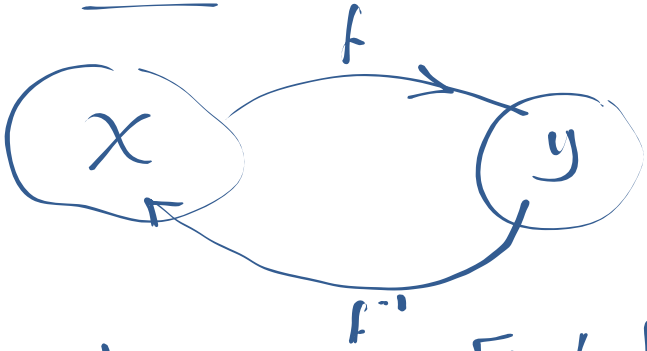
$f \circ g = f(g) = \sqrt{g} = \sqrt{x+1}$

$g \circ f = g(f) = f + 1 = \sqrt{x} + 1$

$f \circ f = f(f) = \sqrt{f} = \sqrt{\sqrt{x}} = \sqrt[4]{x}$

$g \circ g = g(g) = g + 1 = x + 1 + 1$

Inverse Function



$f(x) = x + 5$ Find $f^{-1}(x)$

$y = x + 5 \Rightarrow x = y - 5$

$f^{-1}(x) = x - 5$

$f(x) = 5x^2 + 4$ Find $f^{-1}(x)$

$y = 5x^2 + 4$

$5x^2 = y - 4$

$x^2 = \frac{y-4}{5}$

$x = \sqrt{\frac{y-4}{5}}$

$f^{-1}(x) = \sqrt{\frac{x-4}{5}}$